

Question Paper

Exam Date & Time: 29-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025 / JAN 2025
SUBJECT: MIE 1071/MIE_1071 BASIC MECHANICAL ENGINEERING

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) The steam is initially available at a pressure of 9 bar and dryness 0.98. Find the final quality and temperature of steam at each of the following operations: (4)
- (i) When steam loses 50 kJ/Kg at constant pressure
 - (ii) When steam receives 150 kJ/Kg at constant pressure

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 1B) A chemical company planning to install a coal fired boiler in its plant for process heating is weighing different technical options due to investment constraints. The steam generation capacity required is 5000 kg per hour at a pressure of 0.56 MPa and at a temperature of 220 °C. The feed water is available from a nearby reservoir at an average temperature of 30 °C. The coal consumption is 1500 kg/hr, having a calorific value of 24 MJ/kg. Assess the improvement in boiler efficiency if the following accessories are used. (3)
- i) Economizer which will increase the feed water temperature by 58 °C and reduce the coal consumption by 14%.
 - ii) Air Preheater which will reduce the coal consumption by 20%.
 - iii) Both Economizer and Air Preheater whose combined effect will reduce the coal consumption by 30%.

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 1C) i. Explain the boiler mounting used when the water level in the boiler falls dangerously low. (3)
- ii. A power plant needs a boiler to run its turbines and needs to provide steam generation at a faster rate. Which type of boiler will be most suitable and why?
- iii. Explain the significance of inclined water tubes in Babcock and Wilcox boiler

Course Outcome: "CO3" / Cognitive Level: "Application"

- 2A) A belt drive transmitting power between two pulleys which are 2 meters apart with a speed reduction ratio of 4 has an angle of contact of 3.81 radians. The diameter of the larger pulley is 100 cm and the driver pulley runs at 400 rpm. The initial tension in the belt is 0.95 kN and coefficient of friction is 0.28. Calculate the length of the belt, power transmitted and width of the belt if the permissible tension per meter width of the belt is 10 kN. (4)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 2B) A compound gear train consists of six gear wheels A, B, C, D, E & F. A is the driver gear and F is the driven gear having 100 and 16 teeth respectively. Gear A meshes with gear B having 60 teeth and gear B & C are integral and concentric. Gear C has 80 teeth and meshes with gear D having 100 teeth. Gears D & E are mounted on the same shaft and gear E have 50 teeth meshes with gear (3)

F. Sketch the arrangement of gears and determine the velocity ratio.

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 2C) Suggest and explain with sketch an arrangement when four different speeds are required on the driven shaft. (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3A) Illustrating piston movements explain working of an IC engine that indirectly admits the charge and operates with a compression ratio of 9:1. (4)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 3B) The following data refers to a twin cylinder IC engine generating power in every revolution of the crank shaft. Total stroke volume: 1.5litres, Number of cycles per revolution: 1800, Mean effective pressure: 0.32 MPa, Fuel consumption: 0.00675m³/h, Density of fuel: 800kg/m³, Calorific value of the fuel: 45000kJ/kg. Determine indicated thermal efficiency? (3)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 3C) How do four stroke petrol engines differ from four stroke diesel engines in terms of construction and working? List 6 points. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4A) List and explain any four parts of a Lathe machine. (4)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 4B) With a neat sketch explain the counterboring operation performed on a drilling machine. (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 4C) Explain in detail the working principle of an arc welding process that uses a non consumable electrode. (3)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- 5A) List and explain the common casting defects found in sand casting process. (4)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 5B) Explain how computer control has enhanced productivity and accuracy in CNC machines compared to NC machines. (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 5C) What is additive manufacturing? Explain with suitable example. (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

MIE_1071:BASIC MECHANICAL ENGINEERING

#	Course Outcome	Description	Marks
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1	1	CO 1	17
2	2	CO 2	15
3	3	CO 3	18
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

MIE_1071:BASIC MECHANICAL ENGINEERING

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	3	11(22%)
3	Application	7	22(44%)
4	Analysis	5	17(34%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)